# Worksheet 4 - Limits at Infinity and the Squeeze Theorem

MATH 1271

09/17/2026

## 1 Limits Involving Infinity

1. In this problem, we consider the function $f(x) = \dfrac{4 - x^2}{x^2 + x - 6}$.

   (a) Use algebra to find the value of each of the following limits. If a limit does not exist, write "DNE" and then explain why it does not exist.

   (i) $\lim_{x \to 2} f(x) =$

   (ii) $\lim_{x \to -3} f(x) =$

   (iii) $\lim_{x \to +\infty} f(x) =$

   (iv) $\lim_{x \to -\infty} f(x) =$

   (b) How many vertical asymptotes does $f(x)$ have? Find the equations of all vertical asymptotes of $f(x)$.

   (c) How many horizontal asymptotes does $f(x)$ have? Find the equations of all horizontal asymptotes of $f(x)$.

2. Each of the following limits is some sort of infinity (either $+\infty$ or $-\infty$). Determine which type. Discuss your reasoning with your group.

(a) $\lim_{x \to 5^+} \dfrac{x^3}{x-5}$

(b) $\lim_{x \to 5^-} \dfrac{x^3}{x-5}$

(c) $\lim_{x \to -\infty} \dfrac{x^3}{x-5}$

(d) $\lim_{x \to -2^-} \dfrac{x^2-9}{x^2+5x+6}$

(e) $\lim_{x \to 1^+} \ln(x^2-1)$

**Checking understanding:** Discuss the following questions with your group:

- What if part (d) had asked for $\lim_{x \to -3} \dfrac{x^2-9}{x^2+5x+6}$ instead? How does that change your method and/or final answer?
- What if part (c) had asked for $x \to \infty$ instead? How does that change your method and/or final answer?
- Does it make sense to ask about the one-sided limit $\lim_{x \to 1^-} \ln(x^2-1)$? Why or why not? (Hint: What is the domain of the function $f(x) = \ln(x^2-1)$?)

3. You may assume that the function $y = \dfrac{1-\cos(x)}{x^2}$ satisfies the inequalities

$$\frac{1}{2} - \frac{x^2}{24} < \frac{1-\cos(x)}{x^2} < \frac{1}{2}$$

for all values of $x$ close to $0$. (You can learn where this comes from in Calculus II!) Use this information as needed to find the exact values of each limit below (if a limit does not exist, write "DNE"). If the Squeeze Theorem is used, show all work; if not, then explain why the theorem does not apply.

(a) $\lim\limits_{x\to 0} \dfrac{1-\cos(x)}{x^2}$

(b) $\lim\limits_{x\to \pi} \dfrac{1-\cos(x)}{x^2}$

(c) $\lim\limits_{x\to \infty} \dfrac{1-\cos(x)}{x^2}$

4. Suppose we know that the function $f(x) = \dfrac{\sin(x)}{x}$ satisfies the following inequality when $-\dfrac{\pi}{2} \leq x \leq \dfrac{\pi}{2}$:

$$\cos(x) \leq \frac{\sin(x)}{x} \leq 1$$

(See Apex Calculus Example 1.3.9 for why this holds). Compute the values of the following limits:

(a) $\lim\limits_{x \to 0} \dfrac{\sin(x)}{x}$

(b) $\lim\limits_{x \to 2\pi} \dfrac{\sin(x)}{x}$ (note, the value of this limit is *NOT* $1$).

5. Suppose that a particle is oscillating with position described as a function of time by:

$$s(t) = \sin(t)$$

Use the previous problem to determine the instantaneous velocity of the particle when $t = 0$.